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Accessible delivery storage and anti-entrapment feature for an electric vehicle front trunk

Abstract

A system and method include a first device and a vehicle in communication with the first device. The vehicle includes a front trunk, a camera disposed within the front trunk; and a vehicle controller controlling the camera to capture an image for the front trunk and communicating the image in an image signal to the first device.

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Background/Summary

FIELD

(1) The present disclosure relates generally to electric vehicles and, more particularly, to an electric vehicle having a remotely accessible front trunk (frunk) to prevent entrapment and allow storage for deliveries.

BACKGROUND

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) Electric vehicles typically have a front trunk (frunk) to take the place of the engine compartment. The front trunk is used to secure cargo. The front trunk space of a vehicle can be used to conceal an entrapped human. Reducing the likelihood of a human within the front trunk is desirable. Package delivery services desire more flexible in convenient ways to allow the customers to receive packages. It would be desirable to incorporate the use of the vehicle in the delivery process.

SUMMARY

(4) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(5) The present system incorporates a camera into the front trunk of the vehicle to allow anti human trafficking and enable the front trunk to be used as a delivery locker for a package delivery service.

(6) In one aspect of the disclosure, a system includes a first device and a vehicle in communication with the first device. The vehicle includes a front trunk, a camera disposed within the front trunk; and a vehicle controller controlling the camera to capture an image for the front trunk and communicating the image in an image signal to the first device.

(7) In another aspect of the disclosure, a method of controlling a vehicle having a front trunk includes directing a camera to within the front trunk, controlling the camera to capture an image for the front trunk and communicating the image signal to a first device.

(8) Further areas of applicability will become apparent from the description provided herein. The

description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

(2) FIG. **1** is a perspective view of a vehicle having a front trunk (frunk) with the monitoring system therein.

(3) FIG. **2** is a front view of a camera assembly for use with the monitoring system of FIG. **1**.

(4) FIG. **3** is a block diagrammatic view of the control system for use in FIG. **1**.

(5) FIG. **4**A is a flowchart of a method for authorizing a user.

(6) FIG. **4**B is a flowchart of a method for authorizing a friend to access the frunk.

(7) FIG. **5** is a continuation of both FIGS. **4**A and **4**B to allow a delivery person or another person to access the front.

(8) FIG. **6**A is a screen display for a user device to allow delivery to a particular location such as a frunk.

(9) FIG. **6**B is a screen display for a delivery driver to locate the front.

(10) FIG. **7**A is a screen display for an app for selecting a friend to access the frunk.

(11) FIG. **7**B is a screen display for a friend to access the frunk.

(12) FIG. **8** is a flowchart of a method for enabling someone trapped within the vehicle to be removed.

(13) FIG. **9** is a screen display displayed at the customer device to allow a warning that a person was detected in the frunk displaying a picture and displaying a code to cease the warning when the threat has been removed.

(14) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(15) Example embodiments will now be described more fully with reference to the accompanying drawings.

(16) Referring now to FIG. **1**, a vehicle **10** having a cargo area **11** such as a front trunk **12** (frunk) is illustrated. A front trunk **12** is provided in many battery electric vehicles. The front trunk **12** is enclosed with a hood **14**. The front trunk **12** also has sidewalls **16** and a load floor **18**. The load floor **18** is typically planar and is used to support objects referred to as the load of the vehicle.

(17) The load floor **18**, the sidewalls **16** and the hood **14** form an enclosure for the cargo area **11** that is sealed from the external elements of the vehicle and therefore makes a desirable location for delivery service to deliver a package. Likewise, the cargo area **11** may also be large enough to fit a human.

(18) A camera assembly **20** is disposed having a view of the cargo area **11** within the vehicle. The camera assembly **20** is described in further detail in FIG. **2**.

(19) A latch **22** is used to secure the hood to the vehicle **10** in a closed position. The latch **22** may have a latch actuator which is controlled by a vehicle controller **30**. The vehicle controller **30** may be microprocessor-based and, although represented as one, may comprise a plurality of processors programmed to perform various functions. The vehicle controller **30** may be in communication with the camera assembly **20** and the latch **22** and perform various functions such as unlatching the latch and making control decisions based upon inputs from the camera assembly **20**.

(20) The vehicle **10** may also include lights **34** represented by headlights and a horn **36**, both of which are controlled by the vehicle controller **30**. The lights **34** are headlights that are activated

(flashed) during certain operations described below. In other examples various other lights may be used for signaling alone, or in combination with, the headlights or other lights. Taillights, marker lights, turn signals fog lights are all examples of other lights. The horn **36** may likewise be activated briefly for signaling purposes.

(21) Referring now to FIG. **2**, the camera assembly **20** is illustrated in further detail. In this example, the camera assembly has a plurality of integrated components integrated within a housing **110**. However, although the camera assembly components are illustrated in one housing **110**, discrete components can also be used.

(22) The camera assembly **20** has a camera **112** that has a viewing angle directed into the cargo area **11** of the vehicle **10**. The camera **112** is used to form electronic images from within the cargo area **11**. The camera **112** is capable of taking both still images and sequential images such as in a movie.

(23) The camera assembly **20** may also include an indicator light **114** to provide an indicator to the operator of various functions as described below.

(24) The camera assembly **20** may also include a help button **116**. The help button **116** may be activated and used to trigger a help signal to be communicated to the vehicle controller **30** so that help for a person trapped within the front trunk **12** may be assisted. In response to the help signal the vehicle controller may unlatch a front trunk latch. Details of this are provided below.

(25) The camera assembly **20** also includes a microphone **118** and a speaker **120**. The microphone **118** and the speaker **120** are used to communicate with a person located within the cargo area **11**. The microphone **118** communicates electrical microphone signals to the vehicle controller **30**. The vehicle controller **30** may communicate audible signals to the speaker **120**. The signals to and from the microphone **118** and the speaker **120** may be communicated from the vehicle or from an emergency controller or remote controller as described later in FIG. **3**.

(26) Referring now to FIG. **3**, details of the vehicle controller **30** and the control system **300**, its inputs and outputs, are set forth in further detail.

(27) The camera **112**, the microphone **118**, the speaker **120** and a user interface **310** are in communication with the vehicle controller **30**. The user interface **310** may include the indicator light **114** and the help button **116** described above. The user interface **310** also can include vehicle control buttons **312**. The vehicle control buttons **312** may be located within the frunk **12** may also be distributed within the passenger compartment of the vehicle **10** so that the vehicle **10** operator may provide inputs to the vehicle controller **30**.

(28) The vehicle controller **30** is in communication with a network **314** such as a wireless network. The network **314** is in communication with an emergency controller **316**, a remote controller **318** and a delivery system **320**. The emergency controller **316** may provide emergency services such as emergency services for contacting and dispatching law enforcement. The remote controller **318** may be a centralized controller that is used to communicate through the network **314** to the control module or to the emergency controller **316**. The remote controller **318** may be operated by the manufacturer of the vehicle **10** or a third party. The delivery system **320** is also in communication with the network **314**. The network **314** may interconnect the emergency controller **316**, the remote controller **318** and the delivery system **320**.

(29) The network **314** is also in communication with a customer device **322** that has an (application) app **324** associated therewith. A mobile device, having an app **328**, is also in communication with the network **314**. The customer device **322** and the mobile device **326** together with the apps **324**, **328** may also intercommunicate with the emergency controller **316**, the remote controller **318** and the delivery system **320**.

(30) The customer device **322** corresponds to the device associated with the vehicle **10** and therefore the controller module **30** of a particular vehicle. The customer device **322**, in one example, is a mobile phone. However, other types of customer devices may be used. The mobile device **326** represents various types of devices used by various entities. For example, the mobile

device **326**, in one example, is a mobile device associated with a neighbor or friend that wishes to gain access to the front trunk **12** of the vehicle **10**. A delivery driver may wish to access the front trunk **12** to place a package therein. A friend or neighbor may desire to access the trunk to obtain or borrow a tool or other implement with the permission of the vehicle operator.

(31) The vehicle controller **30** has an antenna **330** for communicating with the network **314**. The antenna **330** may also communicate with a global positioning system (GPS) satellite **332**. The antenna **330** may therefore receive position signals at a global positioning system **340** disposed within the control module. The global positioning system **340** may allow the customer device **322**, the mobile device **326**, the emergency controller **316**, the remote controller **318** and the delivery system **320** to obtain the position of the vehicle by receiving a position signal.

(32) A network interface **342** provides an interface to communicate between the devices **316-326** and the network **314**.

(33) A camera controller **344** is used to control the operation of the camera **112**. The camera controller **344** may also control the microphone **118** and the speaker **120**.

(34) A latch controller **346** is used to control a latch actuator **348** which is used to unlatch the latch **350**. The latch actuator **348**, in one example, is a motor or solenoid used to unlatch the latch **350** to allow the front trunk to open.

(35) A light controller **354** is used to control the lights **34** of the vehicle **10**. For example, the light controller **354** may flash for a delivery driver attempting to deliver a package to the front trunk **12**.

(36) A processor **360** is programmed to perform various control functions as described in greater detail below. The processor **360** is in communication with a memory **362** and a timer **364**. The memory **362** is used to store thresholds, pictures and video by way of example. The timer **364** times various functions. A vehicle display **370** is also controlled by the processor **360**. The vehicle display **370** includes a touch screen or indicator light to indicate the presence of a person or the presence of a package stored within the front trunk **12**.

(37) A transmission positioning sensor **380** provides an electrical signal corresponding to the position of the transmission. In the present example, the transmission signal may indicate whether the transmission is in park or not in park by generating a park signal.

(38) Referring now to FIG. **4**A, steps associated with authorizing another to access the front trunk are set forth. The method sent forth in FIG. **4**A corresponds to when a vehicle owner wants to allow another person such as a friend or relative to pick up something left within the front trunk. For example, if Friend A wants to lend Friend B tools for the weekend and will not be home in person for the exchange, Friend A may place the tools in the frunk and allow access to Friend B to use the code to access the front trunk.

(39) To enable this, step **410** allows the Friend A to select an authorized person, such as Friend B, from a mobile app. The mobile app may have a list of preauthorized users or allows Friend A to enter identifying information such as a telephone number for accessing a text or notification and providing a code thereto. Likewise, an email may also be used to notify Friend B they have access to the frunk of Friend A. In step **412**, a time limit for access by Friend B may be provided. For example, a date and time window that allows one time access may be provided. That is, in combination with step **414**, an access limit may be set. One time access, two time access or unlimited access may be provided. In the above example with respect to tools, two accesses may be provided for Friend B. The friend may be allowed to take out the tools and may be allowed to replace the tools back when they are done by enabling two access. Of course, setting a time limit for access in step **412** and an access limit **414** in step **414** are optional. In step **416**, the enablement of the access (an access signal) is communicated to the user device for Friend B through the network **314** as described above. The remote controller **318** may receive the instructions from the customer device **322** from steps **410-414** and provide an access signal to the mobile device **326**. Examples of screen displays are provided below.

(40) Referring now to FIG. **4**B, the system is also useful for delivering products to a consumer and

using the frunk as a storage locker. In step **430**, the delivery type may be selected by the owner of the vehicle **10** through the use of the customer device **322** using the app **324**. A delivery signal is generated Frunk delivery may be selected and other choices such as the vehicle location **432** may be selected in step **432** and included as part of the delivery signal. The vehicle location **432** may be a general location, such as work, home or school. The high-level or general location will allow the delivery system **320** to schedule an appropriate location. That is, the delivery system **320** may receive the frunk delivery request signal and allow the user to select an address or location for delivery because often times the delivery time is more than a day or two away. The specific parking spot or location is determined on the delivery day.

(41) In step **434**, communication with the delivery system controller **320** is provided. An access signal allowing others to access the vehicle is communicated. The delivery system schedules a delivery for a particular day. Of course, a confirmation of the location of the vehicle **10** closer to the actual delivery time may be communicated through the network **314** to the customer device **322**. In step **436**, the delivery driver may be directed to the general location, such as the work, school or home that the vehicle for receiving the delivery is located. The delivery driver may communicate from the mobile device **326** through the network **314** to the delivery system **320**. The delivery system **320** may communicate with the remote controller **318** and to the vehicle controller **30** to find a more detailed location using the GPS satellite. In step **440**, the specific location such as parking spot may be achieved by communicating the location signal through the network **314** to the mobile device **326** directly or by way of the remote controller of the delivery system **320**.

(42) Referring now to FIG. **5**, after step **416** and after step **440**, access to the front trunk is provided through the system. In step **510**, Friend B or the delivery driver may receive a code to enter into the application **328** of the mobile device **326**. The code may be communicated to the remote controller **318** from the mobile device directly or through the delivery that commands the latch actuator to unlatch the latch **350** using a front trunk unlatch signal. The latch controller **346** may therefore control the latch actuator **348** based upon the entering of the code in step **510** and as commanded by the remote controller **318**. An external input trigger is therefore received at step **512** from the remote controller **318**. In step **516**, it is determined whether the vehicle is in park. In some situations, the vehicle may be occupied or driving or about to be driven. Therefore, a check whether the vehicle is in park is performed in step **516** using the transmission position sensor **380**. If the vehicle is not in park in step **516**, step **518** generates an alert communicated to the customer device **322**, the mobile device or both that access is not available while the vehicle is not in park.

(43) After step **516**, step **520** is performed when the vehicle is in park. In step **520**, a vehicle audio signal using the horn **36** or visual alert signal using the lights **34** (or both) may be provided. As mentioned above, the vehicle lights **34** may be quickly illuminated or the horn **36** may be chirped or both to indicate the location of the vehicle. In step **522**, the frunk latch open sequence may be provided. That is, the latch controller **346** may receive an unlatch control signal from the remote controller **318** in response to a code being entered from the mobile device **326**. The unlatch control signal received at the vehicle controller **30** and the latch actuator **348** releases the latch **350**. As the frunk is opened, step **524** triggers the camera **112**, which is directed to within the frunk, to record an image or a video of the interior or cargo space within the frunk. The controller initiates the capture of the image. The image signal with the image is communicated to a first device such as the vehicle controller, the remote controller or the emergency controller or combinations thereof.

(44) Thereafter, a button may be pushed to close the frunk in step **526**. Likewise, the frunk may also be closed manually. In either situation, the frunk may be locked after being closed by the latch controller **346** controlling the latch actuator **348** to lock the latch **350** by generating a latch signal. Thereafter, a signal may be communicated to the remote controller **318** that the frunk has been closed. That is, in step **528**, a closure signal may be communicated to the remote controller **318**. The remote controller **318** may then allow the code to expire in step **538**. The code may expire when the predetermined number of accesses has been allowed by the vehicle owner. Of course, this

is an optional step since unlimited access may be allowed.

(45) In step **532**, once the latch sequence is initiated, the frunk camera recording or storing of images may be ended. In step **534**, a successful delivery message or a reminder to pick up the package may be communicated to the customer device **322** by way of a text, notification or email.

(46) Referring now to FIG. **6**A, a screen **610** is illustrated for an example of allowing the frunk to be used as a delivery mechanism (a front trunk delivery). General delivery selections **612** are provided. Specifically, the present example provides "home" or "office" as a general delivery area. The home or office location may be stored within the delivery system from prior use or entered upon use for the present delivery. Specific selections **614** are also provided for the location. That is, the specific delivery locations in this example are porch, front desk and frunk. Of course, other locations may be provided. In this example, the frunk is the selection selected as the specific location **614**. The screen display **610** is an example of a screen display displayed by the customer device **322**.

(47) Referring now to FIG. **6**B, a screen display for the delivery driver using the mobile device **326** and the app **328** is set forth. In this example, a map **620** may be displayed with a specific location **622** thereon. The specific location **622** may be a parking lot and may be down to the exact parking spot. When the delivery vehicle or mobile device **326** is closed, the code **624** may be entered for access to the vehicle. The vehicle description displays **626** may provide a general description of the vehicle **10**. When the code is provided, the signal may be communicated from the remote controller **318** to the vehicle controller **30** so that the latch controller **346** unlatches the latch **350** by the latch actuator **348**. Likewise, the light controller **354** may control the lights **24** and the horn **36** to alert the driver as to the specific location of the vehicle.

(48) Referring now to FIG. **7**A, as mentioned above, the system may also be used for allowing a friend to access your frunk. An example of a screen display **710** for selecting a friend for access. In this example, a list **712** may be accessed from an address book or a list stored within the app **324** of the customer device **322**. The list **712** includes Alex, Ben, Charlie and Madelyn. The screen display **710** is displayed at the customer device **322** and to allow a selection box **714** to be moved and for making a selection within the list **712**. In this example, the selection box **714** is provided at the person "Madelyn". Also, an access limit area **716** may also be provided. The access limit area may allow the operator to set an access time and limit for the access. In this example, Aug. 13, 2023, at 10:01 a.m. is the expiration time for the access for Madelyn.

(49) Referring now to FIG. **7**B, a screen display **730** is illustrated. The screen display **730** is illustrated for the mobile device **326**. The screen display **730** may have a message that includes a message **732** for whose trunk, an access limit message **734** for the access limit allowable and a code display **736** for displaying the code. The number of accesses may be displayed at the access number **738**.

(50) Referring now to FIG. **8**, preventing the entrapment of a person within the front trunk may also be performed. In step **810**, the help button **116**, from FIG. **2**, may be triggered when a person is within the vehicle frunk. The help button **116** generates a help signal communicated to the vehicle controller **30**. In step **812**, if the vehicle is in park, a front latch open sequence may be performed to allow the person within the frunk to be removed in step **814**. To unlatch the latch, the help signal from the help button **116** may be communicated to the remote controller **318** which, in turn, communicates with the latch controller **346** through the network **314** to allow the latch actuator **348** to release the latch **350**. Of course, the remote controller **318** may not be involved and therefore the processor **360** may control the opening of the latch **350**. After step **814**, step **816** generates an audio alert or a video alert through the horn **36** or the vehicle lights **34**. In a similar manner, the customer device **322** may generate a screen display corresponding to the successful opening of the front trunk.

(51) Referring back to step **812**, when the vehicle is not in park, step **820** generates an external emergency signal such as a signal to the emergency controller **316** using the 911 emergency

location system in the United States or Canada. That is, when the help button is depressed and the vehicle is in park, as indicated by the transmission position sensor **380**, a position signal with the position of the vehicle as indicated by the global position system **340** may be communicated to the emergency controller **316** along with the help signal to allow emergency services to be deployed or dispatched to the vehicle location. This may be done directly or by the remote controller receiving the help signal and communicating the help signal to the emergency controller. After step **820**, step **822** in response to the help signal, the camera **112** may be triggered to take one or more still pictures or trigger the recording of video from the cargo area within the frunk. The video or pictures may be stored in the memory **362** and/or transmitted to the emergency controller **316**. Likewise, the video may also be accessible by the customer device **322** and communicated thereto.
(52) In step **824**, a mobile alert may be generated by the remote controller **318** and communicated to the customer device **322**. This will allow the customer device to be notified. As mentioned above, the video and/or still pictures may also be communicated. A mobile alert is communicated to the customer device in step **822**. After step **826** and **816**, a user code may be entered in step **828** to end the recording and after the person is removed from the front trunk.
(53) Referring now to FIG. **9**, a screen display **910** is illustrated that generates a warning message **912** indicating a person is detected in the front trunk of the vehicle associated with the owner. A picture **914** of the scene may also be displayed. That is, a picture of the cargo area of the frunk may be displayed. A code entry display **916** may be generated to end the monitoring of the front trunk by entering a code. This is particularly useful to end the monitoring after the person has been removed from the front trunk. Example embodiments are provided so that this disclosure will be thorough and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.
(54) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.
(55) When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.
(56) Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one

element, component, region, layer or section from another region, layer or section. Terms such as "first," "second," and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

(57) Spatially relative terms, such as "inner," "outer," "beneath," "below," "lower," "above," "upper," and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as "below" or "beneath" other elements or features would then be oriented "above" the other elements or features. Thus, the example term "below" can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(58) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A system comprising: a first device; a vehicle in communication with the first device; and a remote controller in communication with the first device and the vehicle, wherein the vehicle includes a front trunk; a camera disposed within the front trunk; and a vehicle controller controlling the camera to capture an image for the front trunk and communicating the image in an image signal to the first device, wherein the first device comprises a user device, wherein the user device generates an access signal that is communicated to the remote controller to allow access to the front trunk; and the remote controller is configured to communicate a code to a mobile device in response to the access signal that permits access to the front trunk; and the remote controller communicates an unlatch control signal to the vehicle controller in response to receiving confirmation of the code from the mobile device, and the vehicle controller controls a front trunk unlatch signal to unlatch the front trunk.
2. The system of claim 1, further comprising a help button disposed within the trunk, said help button generating a help signal and communicating the help signal to the vehicle controller coupled to the camera and triggered to generate the image signal in response to the help signal.
3. The system of claim 2, further comprising an emergency controller receiving the help signal from the vehicle controller and for dispatching help in response to the help signal.
4. The system of claim 3, further comprising a positioning system generating a position signal for the vehicle and wherein the position signal is communicated to the emergency controller in response to the help signal.
5. The system of claim 2, wherein the vehicle further comprises a latch and wherein the latch is unlatched by the vehicle controller in response to the help signal.
6. The system of claim 2, wherein the vehicle further comprises a latch and wherein the latch is unlatched by the vehicle controller in response to the help signal and a park signal indicating the vehicle is in park.
7. The system of claim 1, wherein the remote controller is a delivery system, the access signal comprises a delivery signal, and the user device communicates the delivery signal to the delivery

system corresponding to a front trunk delivery and a vehicle location.
8. The system of claim 7, wherein the delivery system communicates the code to the mobile device.
9. The system of claim 1, wherein the code expires after the unlatch control signal is communicated.
10. A method of controlling a vehicle having a front trunk comprising: communicating a code to a mobile device from a remote controller in response to an access signal; communicating an unlatch control signal to a vehicle controller in response to receiving the code from the mobile device, and controlling a front trunk unlatch signal to unlatch the front trunk using the vehicle controller; directing a camera positioned within the front trunk; controlling the camera to capture an image for the front trunk; and communicating the image signal to a first device.
11. The method of claim 10 further comprising generating a help signal and communicating the help signal to the vehicle controller coupled to the camera and generating the image signal in response to the help signal.
12. The method of claim 11 further comprising receiving the help signal from the vehicle controller at an emergency controller and dispatching help in response to the help signal.
13. The method of claim 12 further comprising generating a position signal for the vehicle at a positioning system and communicating the position signal to the emergency controller in response to the help signal.
14. The method of claim 11 further comprising unlatching a latch by the vehicle controller in response to the help signal.
15. The method of claim 11 further comprising unlatching a latch by the vehicle controller in response to the help signal and a park signal indicating the vehicle is in park.
